

A multi-source calibration framework for activity-based transport simulations integrating background traffic and travel-time perception

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INTRODUCTION

Activity-based multi-agent transport simulations are increasingly used to evaluate urban mobility policies. However, their reliability for policy analysis depends on their ability to reproduce both aggregate mobility behaviour and the resulting traffic conditions.

Existing calibration approaches generally focus on calibration of supply parameters, behavioural parameters, or traffic-count matching using global indicators. However, two aspects that help represent realistic traffic patterns are typically left outside this process. First, synthetic populations generated from household travel surveys represent only resident demand, while the traffic demand of the network also includes freight, service vehicles, and non-resident or tourism-related trips. Second, activity-based simulators such as MATSim assume that agents perceive and evaluate travel times deterministically, whereas real travellers operate under imperfect information and day-to-day variability in experienced conditions, leading to errors in their perceived travel time. Existing calibration approaches leave both of these aspects insufficiently integrated [1, 2]. Consequently, a model may reproduce survey-based mobility indicators while still misrepresenting traffic patterns and experienced travel times.

This study addresses this gap in two steps. Using a large-scale simulation of Paris, we first provide empirical evidence that agreement with household-survey indicators does not guarantee network-level traffic realism. We then develop a multi-source calibration framework combining household travel survey targets, traffic counts, and observed travel times to jointly address behavioural consistency, missing traffic demand, and travel-time perception within a unified calibration process.

METHODOLOGY

Figure 1 presents the proposed calibration framework. After generating the synthetic population using the Eqasim pipeline [1], the framework proceeds through four complementary calibration stages. First, supply calibration corrects the OSM network by adjusting free-flow speeds and link capacities according to lane counts, road types, and fundamental diagrams fitted by link category. Second, behavioural calibration reproduces the EGT 2020 modal shares by distance class by optimizing MATSim scoring parameters using Simultaneous Perturbation Stochastic Approximation (SPSA), followed by a local sensitivity search around the identified optimum. Since SPSA is a heuristic method, its optimization performance may decline after a certain number of iterations, potentially causing the algorithm to stabilize around a local optimum. The subsequent sensitivity analysis therefore helps refine the solution and further improve the calibration results.

Third, background-traffic generation accounts for demand observed by traffic detectors but absent from the resident synthetic population. For each departure hour h , the positive residual between observed and simulated counts, Δ_h , is converted into fixed-route background agents by solving

$$\min_{\mathbf{x}_h \geq 0} \|A_h \mathbf{x}_h - \Delta_h\|_2^2 + \lambda \|\mathbf{x}_h - \mathbf{x}_{0,h}\|_2^2, \quad (1)$$

where A_h is a time-aware route–detector incidence matrix, $\mathbf{x}_{0,h}$ is a prior route-flow distribution, and λ regularizes the solution toward that prior. The estimated flows are introduced as non-adaptive background agents representing traffic components such as freight, services, visitors, and non-resident trips.

Finally, travel-time perception integration relaxes the assumption that agents evaluate travel times without error. Perceived travel time is represented as

$$\tilde{T}_i = T_i(1 + \eta \varepsilon_i), \quad (2)$$

where $\varepsilon_i \sim \mathcal{D}(\phi_{\kappa_i})$ is a dimensionless stochastic perception error fitted empirically using exponentially-modified-Gaussian (EMG) distributions from relative MATSim–TomTom travel-time deviations for each trip context κ_i , and η is a scaling factor selected to minimize route-level error. These context-dependent errors are subsequently incorporated into iterative replanning to introduce realistic heterogeneity in route evaluation.

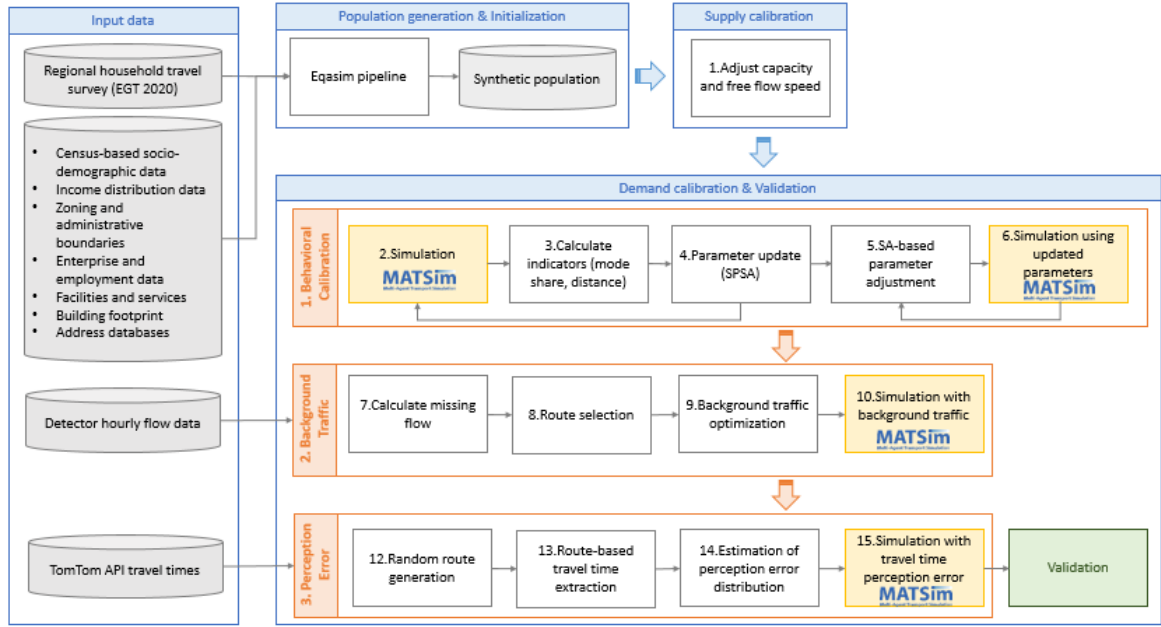


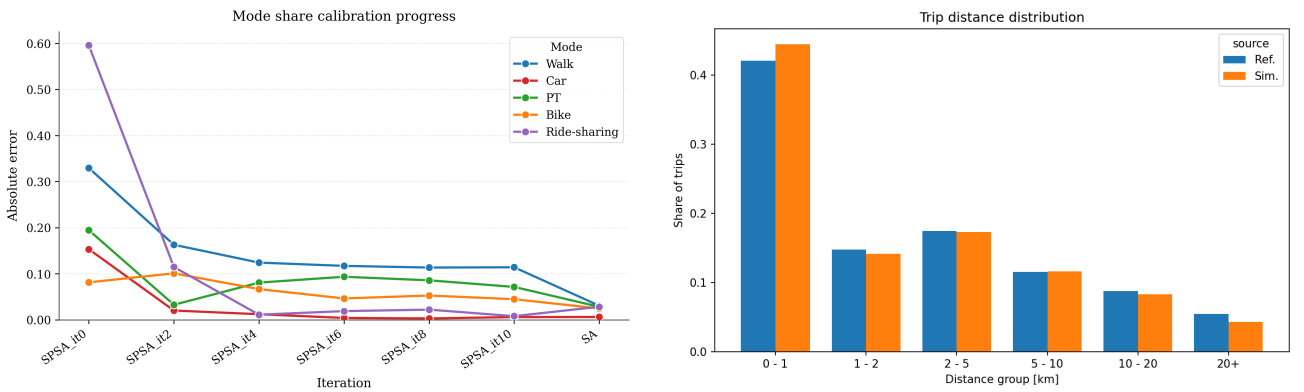
Figure 1. Overview of the four-stage calibration and validation framework combining behavioural, traffic-count, and travel-time data.

RESULTS

The study considers a large-scale MATSim simulation of Paris using a 1% population sample corresponding to 123,178 resident agents. The road network is derived from OpenStreetMap and comprises 654,932 links and 314,735 nodes.

Calibration and validation rely on three complementary data sources including the EGT household travel survey, traffic-count observations, and route-level travel-time data. Hourly traffic counts are obtained from 1,892 loop detectors and averaged over working days in 2019 to construct a representative weekday profile. Travel-time observations are obtained from the TomTom routing API and are used both to evaluate route-level travel-time realism and to estimate travel-time perception-error distributions.

Figure 2 summarizes the behavioural calibration results. Mode share errors decrease during SPSA calibration, while the subsequent local sensitivity search reduces all modal-share errors below 3.1%; the car-mode share is reproduced within 0.6% of the survey target. Simulated trip-distance distributions also remain consistent with the EGT reference distribution.



(a) Mode-share error convergence during SPSA calibration.

(b) Simulated and reference trip-distance distributions.

Figure 2. Validation of aggregate behavioural indicators.

Despite this agreement with survey-based behavioural indicators, the behaviourally calibrated model systematically underestimates observed road traffic. Table 1 compares detector-count fit before and after the integration of background traffic, across daily, peak-period, and hourly aggregation levels. Introducing background traffic substantially improves detector-count agreement across all temporal aggregation levels. Improvements are also observed during both morning and evening peak periods, indicating that the added demand corrects temporally

structured discrepancies rather than simply rescaling daily traffic totals.

These results highlight an important distinction between behavioural consistency and traffic realism in activity-based simulations. Matching modal shares and trip-distance distributions constrains resident travel behaviour, but does not guarantee that the simulated population represents all vehicles observed on the road network.

Table 1. Detector-count fit before and after background-traffic integration across aggregation periods.

Period	Before background traffic			After background traffic		
	R^2	NRMSE	MAE	R^2	NRMSE	MAE
Day	0.897	0.738	7,001	0.946	0.334	3,769
AM peak	0.856	0.599	1,451	0.924	0.379	997
PM peak	0.844	0.552	1,534	0.917	0.371	1,072
Hourly	0.669	0.845	483	0.887	0.454	294

The perception-aware calibration further improves agreement with observed travel times. Table 2 compares travel-time accuracy before and after perception-error integration, evaluated at the network and route level across 832 validated routes. It shows that route-level MAE decreases from 5.92 to 3.61 min. At the network-daily level, MAPE decreases from 28.3% to 21.2%. This pattern suggests that the perception-error model corrects context-dependent biases in individual route evaluation rather than a uniform offset that would show up equally at both scales. Because route choice and downstream behavioural responses in MATSim are driven by these individual-level perceived travel times, this improvement is directly relevant to the credibility of route-choice and mode-choice outcomes, and complements the network-level demand corrections achieved through background-traffic integration.

Table 2. Travel-time accuracy before and after perception-error calibration ($\eta = 3$), evaluated on 832 routes.

Level	Granularity	MAE (min)	RMSE (min)	MAPE (%)
Network	Daily	3.24 $\rightarrow$ 2.42	3.24 $\rightarrow$ 2.42	28.3 $\rightarrow$ 21.2
Network	15-min	3.23 $\rightarrow$ 3.03	4.02 $\rightarrow$ 3.75	32.1 $\rightarrow$ 30.3
Route	15-min	5.92 $\rightarrow$ 3.61	10.07 $\rightarrow$ 5.99	66.4 $\rightarrow$ 63.5

CONCLUSION

Calibration of activity-based transport simulations should not be limited to reproducing aggregate behavioural indicators. In the Paris case study, satisfactory agreement with survey-based modal shares and trip distance distributions coexists with substantial underestimation of observed traffic, demonstrating that behavioural realism is necessary but not sufficient for traffic representation. Introducing missing background demand improves traffic count agreement, while incorporating stochastic travel time perception substantially reduces route level travel time errors. The proposed multi-source framework provides a more interpretable calibration process and reduces the risk of compensating for missing demand or network discrepancies through behavioural parameters alone. This provides a stronger basis for subsequent policy and “what-if” analyses in large-scale MATSim applications.

The authors intend to present this work in person at MUM26.

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